Week 4 – Deployment on Flask (Report)

Name: Magdalena Ivanova
Batch code: LISUM49
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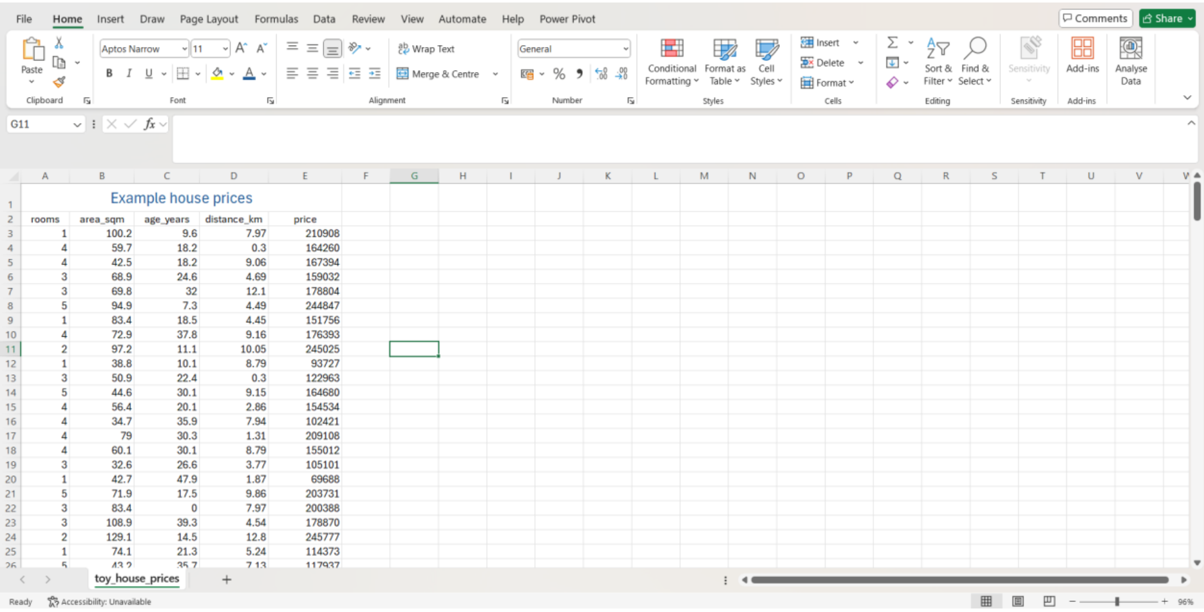
1. Toy Data (CSV)

Dataset: data/example_house_prices.csv

Short description:

- synthetic housing dataset with columns: rooms, area_sqm, age_years, distance_km, price



Example house prices

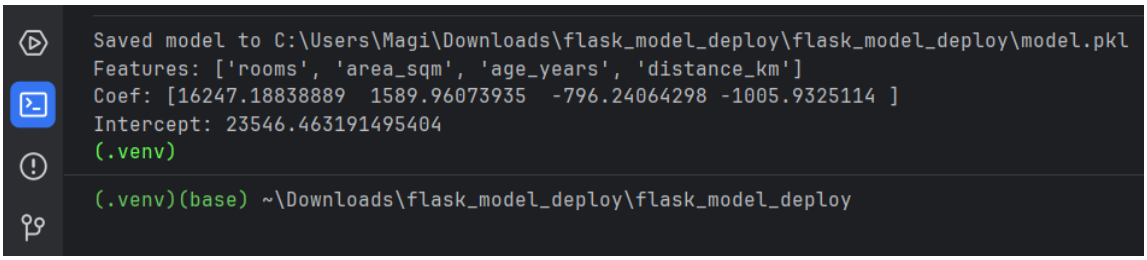
rooms	area_sqm	age_years	distance_km	price
1	100.2	9.6	7.97	210908
4	59.7	18.2	0.3	164260
4	42.5	18.2	9.06	167394
3	68.9	24.6	4.69	159032
3	69.8	32	12.1	178804
5	94.9	7.3	4.49	244847
1	83.4	18.5	4.45	151756
4	72.9	37.8	9.16	176393
2	97.2	11.1	10.05	245025
1	38.8	10.1	8.79	93727
3	50.9	22.4	0.3	122963
5	44.6	30.1	9.15	164680
4	56.4	20.1	2.86	154534
4	34.7	35.9	7.94	102421
4	79	30.3	1.31	209108
4	60.1	30.1	8.79	155012
3	32.6	26.6	3.77	105101
1	42.7	47.9	1.87	69688
5	71.9	17.5	9.86	203731
3	83.4	0	7.97	200388
3	108.9	39.3	4.54	178870
2	129.1	14.5	12.8	245777
1	74.1	21.3	5.24	114373
5	43.2	35.7	7.13	117937

2. Model Training

Method: NumPy linear regression (normal equation)

Command used:

- "python train_model.py"



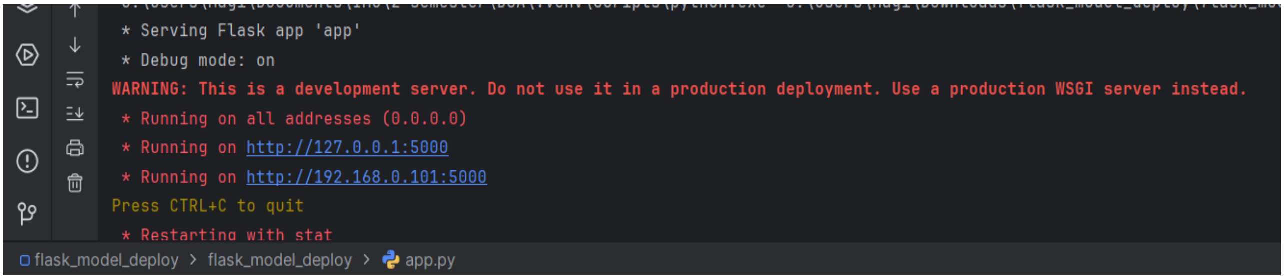
```
Saved model to C:\Users\Magi\Downloads\flask_model_deploy\flask_model_deploy\model.pkl
Features: ['rooms', 'area_sqm', 'age_years', 'distance_km']
Coef: [16247.18838889  1589.96073935  -796.24064298 -1005.9325114 ]
Intercept: 23546.463191495404
(.venv)

(.venv)(base) ~\Downloads\flask_model_deploy\flask_model_deploy
```

3. Flask App Deployment

Command used:

- “pip install -r requirements.txt”
- “python app.py”

App URL: `http://127.0.0.1:5000` (or `http://127.0.0.1:8080` if you changed PORT)



4. Prediction Results

Form input example: rooms=6, area_sqm=250, age_years=1, distance_km=3

Result: Predicted price shown on the screenshot below